

Module 3 – Lab 2

Regression by Linear Algebra

- You may use Excel or R to complete this lab.
- In this lab, we will perform linear regression by using linear algebra. A set of paired data is given. Use linear algebra to:
 1. Perform Simple Regression (Use the Simple Regression Worksheet)
 - (i) Calculate the slope and the intercept of the regression line (formula 6 below).
 - (ii) Determine the influence matrix and use it to calculate the predicted y values (formula 8).
 - (iii) Calculate the residuals directly by using the influence matrix (formula 9).
 - (iv) Calculate the mean squared error MSE by using the influence matrix (formula 10).
 - (v) Compare your calculated values in parts 1-4 above with those obtained when we do not use linear algebra.
 2. Perform Multiple Linear Regression (Use the Multiple Regression Worksheet).
Complete parts (i)-(v) as described for the simple regression case.

Theoretical Backgrounds:

A little application of linear algebra will allow us to avoid a lot of the book-keeping details and will make multiple linear regression as simple as simple linear regression.

Model:

Suppose that our data consists of n pairs of observations (x_i, y_i) , $i = 1, 2, \dots, n$; of the predictor variable X and the response variable Y . In a simple linear regression model, we wish to fit the following model to the data:

$$Y = \beta_0 + \beta_1 X + \epsilon, \quad (1)$$

where ϵ is the residuals of the model, $E(\epsilon) = 0$, and $VAR(\epsilon) = \sigma^2$ (a constant). Furthermore, we assume that ϵ is uncorrelated across the measurements.

Methodology:

Let's consider the two coefficients of the model as a 2×1 vector β ; i.e., $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$. Furthermore, we group the observations of the response variable Y as a $n \times 1$ vector y . On the other hand, the observations of the predictor variable X are grouped as a $n \times 2$ matrix x as shown below

(this may look a little unusual at first, but we will see why we prefer to express the predictor variable in this way).

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \quad (2)$$

Note what will happen when we multiply the matrix $\mathbf{x}$ by the vector $\boldsymbol{\beta}$:

$$\mathbf{x} \boldsymbol{\beta} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 x_1 \\ \beta_0 + \beta_1 x_2 \\ \vdots \\ \beta_0 + \beta_1 x_n \end{bmatrix}. \quad (3)$$

That is, $\mathbf{x} \boldsymbol{\beta}$ is the $n \times 1$ matrix which contains the point predictions. The matrix $\mathbf{x}$ is called the *design matrix* for the regression model.

Therefore, the vector of the residuals (i.e., the errors) can be expressed in the vector form as:

$$\mathbf{e} = \mathbf{y} - \mathbf{x} \boldsymbol{\beta} \quad (4)$$

The total squared error $\mathbf{SE}$ is then the inner-product of $\mathbf{e}$ with itself; i.e., $\mathbf{SE} = \mathbf{e}^T \mathbf{e}$, and the mean squared error $\mathbf{MSE}$ is the average of total squared error across all data points.

$$\mathbf{MSE} = \frac{1}{n} \mathbf{e}^T \mathbf{e} \quad (5)$$

In the theory of linear regression, the estimated coefficients of the regression ($\hat{\boldsymbol{\beta}}$) are selected so that $\mathbf{MSE}$ attains a minimum possible value. It turns out that the solutions of this optimization problem are:

$$\hat{\boldsymbol{\beta}} = (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \mathbf{y} \quad (6)$$

Therefore, the predicted values are calculated by:

$$\hat{\mathbf{y}} = \mathbf{x} \hat{\boldsymbol{\beta}} = \mathbf{x} (\mathbf{x}^T \mathbf{x})^{-1} \mathbf{x}^T \mathbf{y}$$

$$\hat{\mathbf{y}} = \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (7)$$

In this formula, the matrix $\mathbf{H} = \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ is called the **Influence Matrix**. The formula for the predicted values can then be expressed in terms of the influence matrix $\mathbf{H}$.

$$\hat{\mathbf{y}} = \mathbf{H} \mathbf{y} \quad (8)$$

Note that $\mathbf{H}$ does not depend on $\mathbf{y}$. Furthermore, it is a symmetric matrix; that is, $\mathbf{H}^T = \mathbf{H}$.

The matrix $\mathbf{H}$ plays an important role in modeling linear regression. We note that the residuals can be expressed in terms of $\mathbf{H}$; thus can be calculated directly by the formula:

$$\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H}) \mathbf{y}$$

$$\mathbf{e} = (\mathbf{I} - \mathbf{H}) \mathbf{y} \quad (9)$$

Here, $\mathbf{I}$ is the identity matrix whose dimensions are the same as those of $\mathbf{H}$.

In fact, the mean squared error (MSE) can also be calculated directly by using the influence matrix $\mathbf{H}$.

$$\text{MSE} = \frac{1}{n} \mathbf{e}^T \mathbf{e} = \frac{1}{n} \mathbf{y}^T (\mathbf{I} - \mathbf{H})^T (\mathbf{I} - \mathbf{H}) \mathbf{y}$$

$$\text{MSE} = \frac{1}{n} \mathbf{y}^T (\mathbf{I} - \mathbf{H})^T (\mathbf{I} - \mathbf{H}) \mathbf{y} \quad (10)$$